

Dane Dathrego

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TECHNICAL SKILLS

Languages: C/C++, Python, SQL, SystemVerilog, JavaScript, Java

Frameworks & Platforms: React, Node, Express, Jest, MongoDB, Git, Xilinx Vivado

Embedded & Hardware: SPI, I2C, UART, GPIO, Oscilloscopes, Logic Analyzers, Soldering

Domains: Full-Stack Development, Embedded Systems, Robotics, Computer Networking, Database Management, Agile Development, Signal Processing, Electrical Circuits, Operating Systems

PROJECTS

Poly1rover Project Port II — *CPE Capstone and Senior Project* September 2025 - Present

- Achieved reliable autonomous test tube retrieval by integrating the AI vision and kinematic arm engines in Python/C, enabling the vision system to pass real-time coordinates to the arm for complete end-to-end pickup sequences
- Improved detection accuracy through iterative camera calibration and coordinate translation refinements, and resolved hardware issues including motor wiring, polarity, and port failures through repeated testing cycles
- Advanced wireless communication by replacing a USB Wi-Fi dongle with an SPI-based WFM200S chip, implementing a custom Vivado hardware design and backporting a Linux kernel driver to free the USB port for camera operation

What's For Dinner? — *Software Engineering* January 2025 - June 2025

- Built a full-stack recipe recommendation web app across a two-quarter Agile cycle, contributing to all phases from requirements gathering and UI/UX design to deployment, using React, Node/Express, and MongoDB
- Reduced manual testing overhead by implementing automated Jest test suites, supporting continuous deployment and feature iteration throughout the project lifecycle

Frequency-Controlled Actuator — *Electrical and Electronic Circuits* April 2024 - June 2024

- Designed and built a modular signal-processing circuit that drives a solenoid based on real-time audio frequency analysis, validating signal integrity and actuator response using oscilloscopes, bench power supplies, and multimeters

EDUCATION

California Polytechnic State University, San Luis Obispo, CA September 2022 - June 2026 (Expected) — *B.S. Computer Engineering, Minor in History*

- Fourth year undergraduate with a 3.6 GPA
- Member of the Computer Engineering Society, Cal Poly CS+AI, and Poly1 Rover

EXPERIENCE

Cal Poly ITS Client Services, San Luis Obispo, CA — *Student Assistant* January 2026 - Present

- Resolved hardware, software, and network issues for faculty, staff, and students across Windows, macOS, iOS, and Android platforms, maintaining accurate ticket documentation in Jira
- Improved deployment efficiency by configuring, imaging, and deploying computers and peripherals while translating technical solutions into clear guidance for non-technical users

e2f, Los Altos, CA (Remote) — *AI Research Intern* July 2025 - September 2025

- Supported model training and evaluation pipelines by delivering high-accuracy data annotation and quality review, directly contributing to benchmarking outcomes
- Accelerated internal research decisions by producing detailed analytical reports on machine learning systems, informing product development direction across multiple projects
- Gained hands-on exposure to the full AI model lifecycle within a commercial research environment, applying findings directly to ongoing product work